

Max Shen

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EDUCATION

University of Southern California , Los Angeles, CA <i>M.S. in Computer Science</i>	Jan 2026 – May 2027 <i>GPA: 4.0 / 4.0</i>
National Chin-Yi University of Technology , Taichung, Taiwan <i>B.Eng. in Computer Science and Information Engineering</i>	Sept 2019 – Jun 2023

RESEARCH EXPERIENCE

SLURM Lab , USC <i>Advisor: Daniel Seita</i>	Jan 2026 – Present <i>Los Angeles, CA</i>
Formulating uncertainty estimation to benchmark robotic world models and flag inaccurate policy rollouts, validated on a physical DROID Franka platform.	
Smart Manufacturing Lab , NCUT <i>Advisor: Yu-Chi Liu</i>	Aug 2022 – Jun 2023 <i>Taichung, Taiwan</i>
<i>Project: Adaptive Thermal Equilibrium Control System for CNC Machines</i>	
- Formulated a closed-loop framework coupling an MLP with MOPSO to predict and actively compensate for spindle thermal tilting ($\Delta X, \Delta Y$) to within $0.5\mu\text{m}$ RMSE on held-out test data. - Suppressed thermal drift from $3\text{--}12\mu\text{m}$ to under $1\mu\text{m}$ on a physical 3-axis CNC vertical machining center.	

PROFESSIONAL EXPERIENCE

Graphen – R&D Team <i>Robotics AI Intern</i>	May 2026 – Aug 2026 <i>Taipei, Taiwan</i>
- Scaled a single-arm platform to synchronized long-horizon (40s+) bimanual manipulation, fine-tuning NVIDIA Isaac GR00T via Sim-to-Real imitation learning to lift task success from 20% to 60% in simulation. - Built a dual real-and-simulation demonstration pipeline – teleoperation and MoCap capture on hardware, Isaac Sim collection under Isaac Lab domain randomization – doubling collection to 50 episodes/hr behind automated QA that rejects outlier, frozen, and truncated episodes. - Trained a Diffusion Transformer manipulation policy with distributed multi-GPU PyTorch, augmenting demonstrations via NVIDIA Cosmos synthetic generation, reaching 60% OOD success under real-robot evaluation.	
DoubleSpot <i>Software Engineer Intern</i>	Sep 2025 – May 2026 <i>Seattle, WA</i>
- Partnered with the start-up CTO to decompose a coupled legacy monolith into decoupled React and Express services along domain-driven boundaries, halving feature delivery cycles from two months to one. - Codified AWS infrastructure as Terraform (ECS, ALB, RDS, S3) behind an end-to-end CI/CD pipeline, replacing manual deployment with automated build, test, and container rollout in 15–20 minutes. - Built a multi-tenant React Native application enforcing row-level tenant isolation across the state-management and API-authorization layers, onboarding 20+ enterprise clients.	
Yois <i>Software Engineer</i>	Aug 2023 – Jul 2025 <i>Taipei, Taiwan</i>
- Kept inventory consistent across 10+ distributed store nodes by designing scheduled batch reconciliation between in-store POS and the backend e-commerce database, cutting overselling to 0.05%. - Led a 5-engineer database migration, cutting read latency 40% and absorbing peak promotional traffic by refactoring slow RDBMS queries, backfilling indexes, and caching hot-product and inventory lookups in Redis. - Migrated the backend to AWS (EC2, RDS, ELB) and gated releases behind CI/CD with static analysis, unit tests, and automated rollback, cutting verification from 2–4 weeks of manual testing to one.	

SKILLS

Programming Languages: Python, C++, TypeScript, JavaScript, Java, SQL
AI & Robotics: PyTorch, ROS2, Isaac Sim/Lab, LeRobot, TensorFlow, OpenCV
Web & Infrastructure: React, React Native, Express, FastAPI, Docker, AWS, Terraform, CI/CD, Linux, Git